



Contact



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The Hong Kong University of Science
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Clear Water Bay, Kowloon
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[.yasaman--mohtat](#)

Education

MPhil in Civil and Environmental Engineering

(Sep 2026 – Present)

Department of Civil and
Environmental Engineering, HKUST
Supervisor: Prof. Samir Kumar Khanal

Master of Mining and Environment (2024 – August 2026)

School of Mining Engineering, Faculty
of Engineering, University of Tehran
GPA: 4/4 – 18.14/20

Bachelor of Science in Mining Engineering

(2019 – January 2024)

School of Mining Engineering, Faculty
of Engineering, University of Tehran

Software Skills

- ArcGIS, Arcmap, Oasis Montaj
- Datamine, AutoCad
- SEEP/W, CTRAN/W, Comsol
- Python, C++, SPSS, Design Expert

Curriculum Vitae

Yasaman Mohtat

Graduate student in Civil and Environmental Engineering at HKUST, with a keen interest in phytoremediation and environmental geochemistry, centered on the removal of heavy metals and trace elements from contaminated soil and water. Previously completed an M.Sc. in Mining and Environment at the University of Tehran on trace element leaching from the low-grade ore of the Miduk copper mine.

Professional Experience

September 2024 – August 2026

Laboratory Expert

School of Mining Engineering, University of Tehran, Tehran

- Conducting environmental and geochemical laboratory analyses related to mining projects
- Preparing and analyzing soil and rock samples for heavy metal and mineral content
- Operating and maintaining laboratory instruments (AAS, ICP, and spectrophotometers)

August 2023 – September 2024

Mining Contractor

Kani Azarin Pima Company

- Project discovery: monitoring tender information databases
- Identifying cooperation opportunities with public and private companies and organizations

June 2023 – August 2023

Soil Mechanics Laboratory Intern

Technical and Soil Mechanics Laboratory, Tehran, Iran

- Granulation test
- Conducting aggregate impact value test
- Conducting frost test

September 2022 – June 2023

Laboratory Intern in Processing and Geochemistry

School of Mining Engineering, University of Tehran, Iran

- Conducting flotation tests
- Sample identification and chemical analysis
- Conducting X-ray analysis
- Determining the work index (Asia bullet bond)

Research Areas

- Acid mine drainage (AMD) prevention and treatment
- Soil and groundwater contamination control
- Environmental geochemistry
- Air pollution exposure assessment and mitigation
- Mineral–water interactions and heavy metal mobility

Publications

1. A Novel Fuzzy Deep Groundwater Sustainability Index FDGSI for Climate-Smart Mining and Sustainability-Aimed Governance. Hamid Sarkheil 1*, Emad Rostamian 2, Shahrokh Rahbari 3, **Yasaman Mohtat 4**, Razieh Lak 5 (**Submitted**)
2. Investigation and Modelling of Trace-Element Leaching from the Low-Grade Ore of the Miduk Porphyry Copper Mine: An Integrated Experimental, Kinetic, Machine-Learning and Petrographic Image-Analysis Study. **Yasaman Mohtat a,***, Faramarz Doulati Ardejani a, Ali Najmeddin a, Kiarash Yazdani b, Hiva Farhadi a (**Preparing**)
3. Sarkheil, H. Shahbaznejad, M. Rayegani, B. **Mohtat, Y.** Salahjou, T. Sadeghy Nejad, A. Lithium extraction assessment from brines in Kerman province: challenges and opportunities for clean energy transition and climate change mitigation (2025).
4. **Mohtat, Y.** & Doulati Ardejani, F. Investigation and Modeling of Trace Element Leaching from Low-Grade Ore of Miduk Copper Mine. [thesis submitted for Master of Science in Mining Engineering, Mining and Environment specialization, University of Tehran] (2026).
5. **Mohtat, Y.** & Noaparast, M. Enriching lead and gold-bearing zinc ore's sample: Nadushan mine, Yazd, Iran. [thesis submitted for Bachelor of Science in Mining Engineering, University of Tehran] (2023).
6. Mohseni, V., **Mohtat, Y.**, Rezaei, A. & Noaparast, M. Environmental effects caused by flotation and production of lead concentrate. The third conference on mining and green mining industries of Iran, 2023, Zanjan, Iran.

Awards and Honors

1. Member, Society for Mining, Metallurgy & Exploration (SME)
2. Achieved second-highest GPA among postgraduate students in the Mining and Environment program
3. Student at the University of Tehran (The first ranked university in Iran)
4. Among the top 20% of Mining Engineering students in the 2019 undergraduate cohort
5. High-school graduate from Sampad (National Organization for Development of Exceptional Talents)
6. Ranked 13th nationwide in the M.Sc. entrance exam

Selected Projects

Graduate Selected Projects

(M.Sc in Mining and Environment)

- Analysis of mining regulations and environmental protection policies in protected areas.
- Translation, synthesis, and critical summary of international research articles related to mining and environmental management.
- Preparation of spatial distribution maps for environmental contamination (GIS-based mapping) for academic publication.
- Collaborative research with Imam Khomeini International University (IKIU), Qazvin, on sustainable mining practices and environmental protection.

Undergraduate Selected Projects

(B.Sc in Mining Engineering)

- Examining the production line of the ore processing plant
- Investigating the control and maintenance system of Angouran mine, Sheykhlar, Mahneshan County
- Exploring the mineral deposits of Kahak, Salafchegan, Iran
- Investigating the causes and methods of coal washing
- Checking the ventilation system of Iran Manganese Mine (Venarch), Qom, Iran
- Exploring the importance and role of mining
- UCS test
- Developing methodologies for the regional exploration of minerals
- Determining elastic modulus and Poisson's ratio